SCM LAB REPORT

Introduction

1.What is Git ?
Ans :- Git is a distributed version control system used to track changes in source code during software devlopment . It helps multiple developers collaborate efficiently.

2.What is version control?
Ans :-Version control is a system that records changes to flies over time. It allows developers to revert to previous versions and collaborate without conflicts.

3.What is Github?
Ans :- Github is a cloud-based platform that hosts Git repositories and enables collaboration ,version tracking and project management.

4.Difference between Git and GitHub?
Ans :-
---------- Git ----------
***Git is a tool .**
***Git works locally.**
***Git manages version control.**
***Git does not require internet.**
***Git is software.**
---------- GitHub ----------
***GitHub is a platform.**
*** GitHub works online.**
***GitHub hosts repositories.**
***GitHub requires internet.**
***GitHub is a service.**

Commands with description

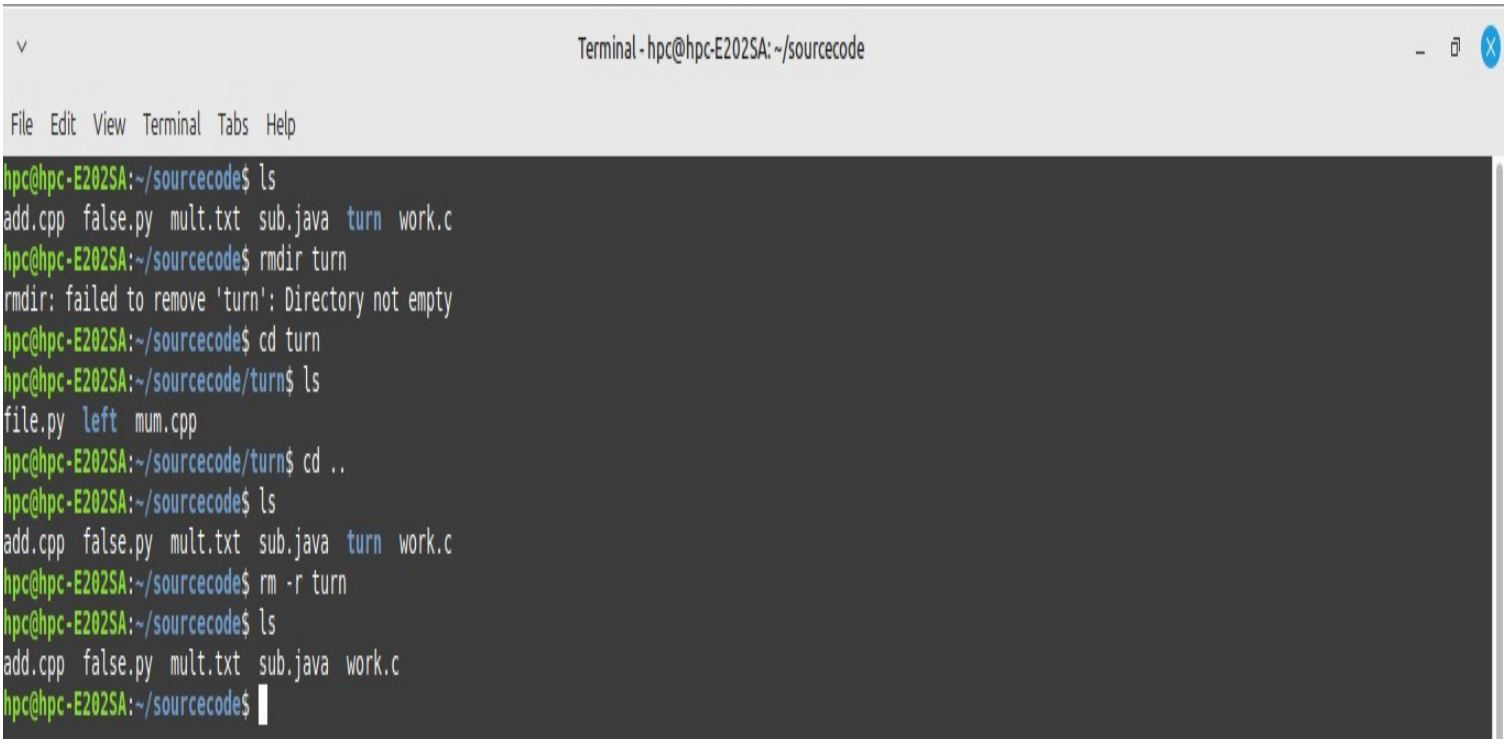
- **git –version :- Checks the installed Git version on our system.**
- **git config –list :- Shows all Git configuration settings.**
- **pwd :- Shows the current working directory (present folder location).**
- **cd folder_name/location :- Changes the current directory to another folder.**
- **ls :- Displays all files and folders in the current directory.**
- **touch filename :- Ceates new empty file.**
- **rm filename :- Deletes a file.**

```
Terminal - hpc@hpc-E202SA: ~/sourcecode
File  Edit  View  Terminal  Tabs  Help
hpc@hpc-E202SA:~$ git --version
git version 2.43.0
hpc@hpc-E202SA:~$ git config --list
user.name=srishanth1211
user.email=p.srishanth@s.amity.edu
hpc@hpc-E202SA:~$ pwd
/home/hpc
hpc@hpc-E202SA:~$ cd sourcecode
hpc@hpc-E202SA:~/sourcecode$ ls
hpc@hpc-E202SA:~/sourcecode$ touch new.py
hpc@hpc-E202SA:~/sourcecode$ ls
new.py
hpc@hpc-E202SA:~/sourcecode$ rm new.py
hpc@hpc-E202SA:~/sourcecode$ ls
hpc@hpc-E202SA:~/sourcecode$ touch add.py
hpc@hpc-E202SA:~/sourcecode$ touch sub.java
hpc@hpc-E202SA:~/sourcecode$ touch mult.txt
hpc@hpc-E202SA:~/sourcecode$ touch divd.py
hpc@hpc-E202SA:~/sourcecode$ touch true.py
hpc@hpc-E202SA:~/sourcecode$ ls
add.py  divd.py  mult.txt  sub.java  true.py
hpc@hpc-E202SA:~/sourcecode$ touch false.c
hpc@hpc-E202SA:~/sourcecode$ touch hdd.cpp
hpc@hpc-E202SA:~/sourcecode$ touch make.java
hpc@hpc-E202SA:~/sourcecode$ touch work.c
```

- **mkdir folder_name :- Creates a new folder.**
- **rmdir folder_name :- Deletes an empty file.**

```
Terminal - hpc@hpc-E202SA: ~/sourcecode
File  Edit  View  Terminal  Tabs  Help
hpc@hpc-E202SA:~/sourcecode$ mkdir scf
hpc@hpc-E202SA:~/sourcecode$ mkdir thar
hpc@hpc-E202SA:~/sourcecode$ mkdir lal
hpc@hpc-E202SA:~/sourcecode$ mkdir singh
hpc@hpc-E202SA:~/sourcecode$ mkdir india
hpc@hpc-E202SA:~/sourcecode$ ls
add.cpp  false.py  india  lal  mult.txt  scf  singh  sub.java  thar  work.c
hpc@hpc-E202SA:~/sourcecode$ rmdir india
hpc@hpc-E202SA:~/sourcecode$ rmdir lal
hpc@hpc-E202SA:~/sourcecode$ rmdir scf
hpc@hpc-E202SA:~/sourcecode$ rmdir singh
hpc@hpc-E202SA:~/sourcecode$ rmdir thar
hpc@hpc-E202SA:~/sourcecode$ ls
add.cpp  false.py  mult.txt  sub.java  work.c
hpc@hpc-E202SA:~/sourcecode$
```

- **cd .. :- it is used to move one directory back (go to parent diectory).**
- **Rm -r foldername :- Deletes a folder and its contents(non-empty folder).**



- **cat filename.txt :- Displays content of a file.**
- **vi filename.txt :- Opens a file in vi editor(text editor in terminal).**
- **echo “text” > :- Write text into a file.**
- **echo “text” >> :- Append text into a file(ad content).**

Terminal - hpc@hpc-E202SA: ~/sourcecode

File Edit View Terminal Tabs Help

```
hpc@hpc-E202SA:~$ cd sourcecode
hpc@hpc-E202SA:~/sourcecode$ ls
add.cpp  false.py  mult.txt  sub.java  work.c
hpc@hpc-E202SA:~/sourcecode$ cat mult.txt
hpc@hpc-E202SA:~/sourcecode$ vi mult.txt
hpc@hpc-E202SA:~/sourcecode$ echo "HELLO" > sri.txt
hpc@hpc-E202SA:~/sourcecode$ echo "WORLD" >> sri.txt
hpc@hpc-E202SA:~/sourcecode$ ls
add.cpp  false.py  mult.txt  sri.txt  sub.java  work.c
hpc@hpc-E202SA:~/sourcecode$
```

sri.txt (~/sourcecode)

File Edit View Search Tools Documents Help

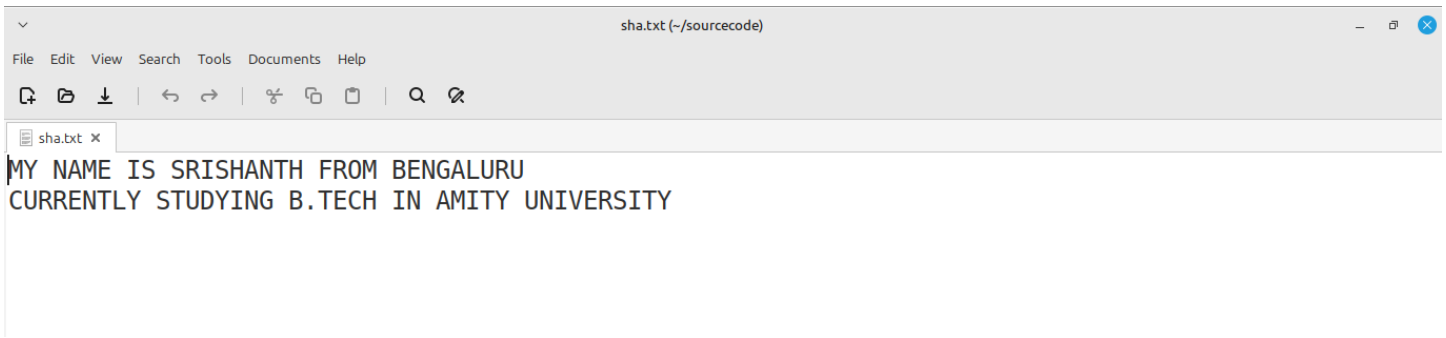
sri.txt

```
HELLO
WORLD
```

Terminal - hpc@hpc-E202SA: ~/sourcecode

File Edit View Terminal Tabs Help

```
hpc@hpc-E202SA:~$ cd sourcecode
hpc@hpc-E202SA:~/sourcecode$ cat > sha.txt
MY NAME IS SRISHANTH FROM BENGALURU
hpc@hpc-E202SA:~/sourcecode$ cat >> sha.txt
CURRENTLY STUDYING B.TECH IN AMITY UNIVERSITY
hpc@hpc-E202SA:~/sourcecode$
```



- **cat file1> file2 :- Overwriting the existing content.**

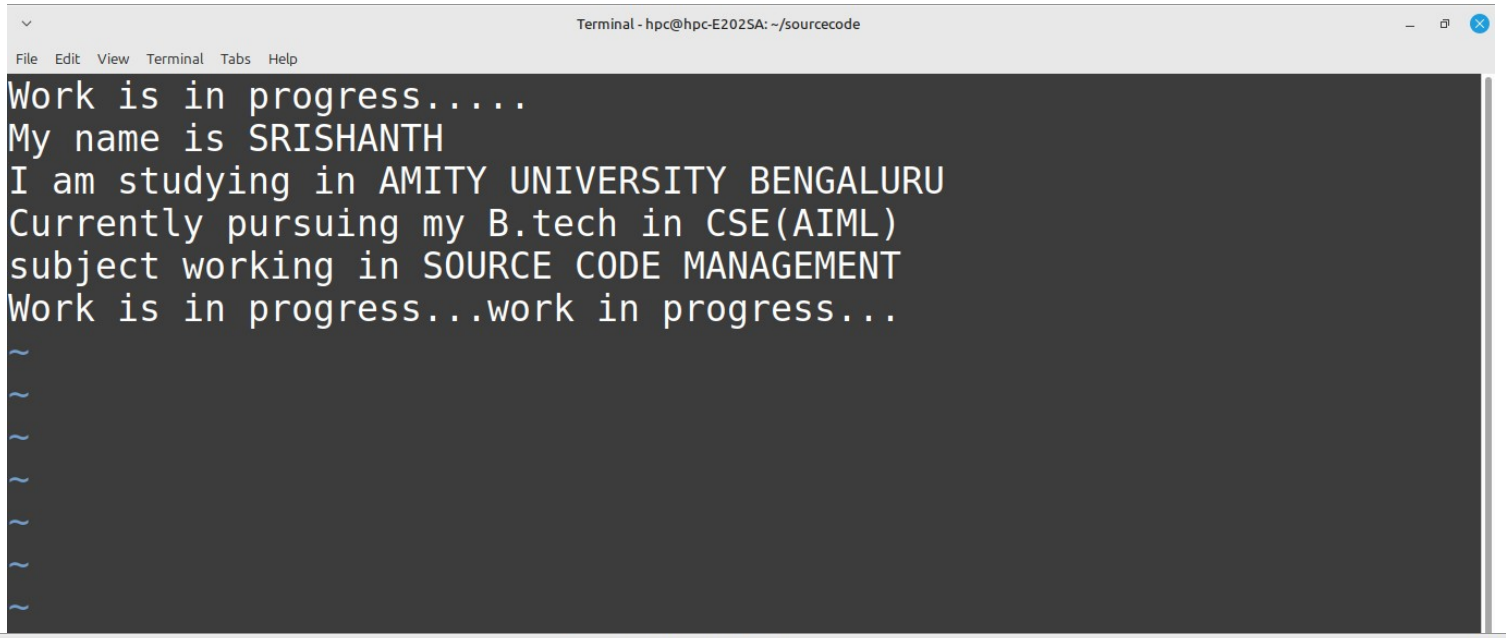
- **nano filename :- is a simple text editor used to create and edit files in the terminal.**

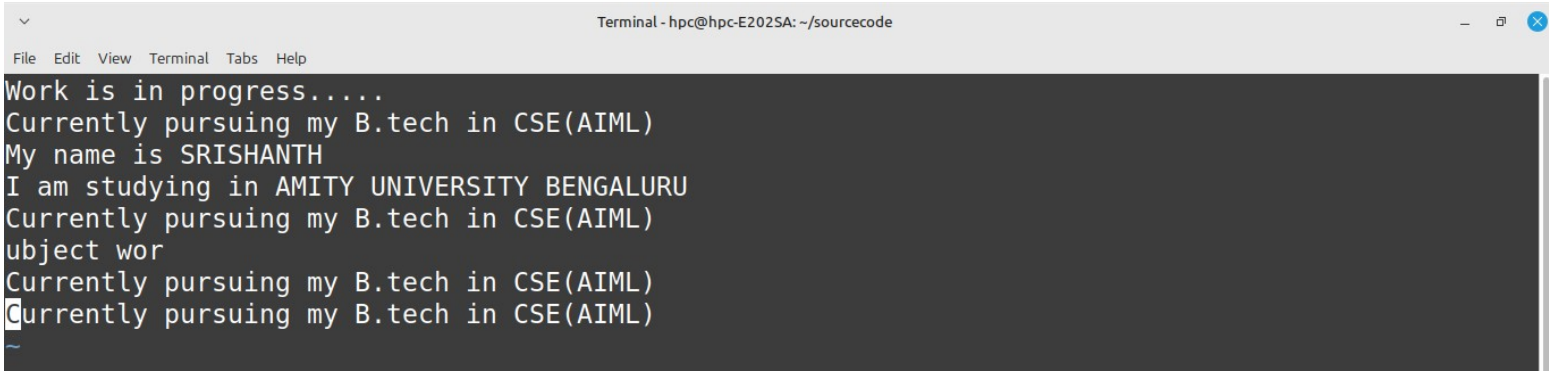
```
Terminal - hpc@hpc-E202SA: ~/sourcecode
File  Edit  View  Terminal  Tabs  Help
  GNU nano 7.2                          ram.txt *
I AM IN INDIA...

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

```
Terminal - hpc@hpc-E202SA: ~/sourcecode
File  Edit  View  Terminal  Tabs  Help
hpc@hpc-E202SA:~$ cd sourcecode
hpc@hpc-E202SA:~/sourcecode$ echo "HELLO SOURCE" > work.txt
hpc@hpc-E202SA:~/sourcecode$ echo "Learning vi editor" >> work.txt
hpc@hpc-E202SA:~/sourcecode$ cat work.txt
HELLO SOURCE
Learning vi editor
hpc@hpc-E202SA:~/sourcecode$ cat > work2.txt
I AM IN INDIA...
HELLO
WORLD
hpc@hpc-E202SA:~/sourcecode$ cat work2.txt
I AM IN INDIA...
HELLO
WORLD
hpc@hpc-E202SA:~/sourcecode$
```

- **Deletes one character -> x**
- **Deletes one full line -> dd**
- **Copy a line -> yy**
- **Paste it twice -> p**
- **Paste above ->P**





----------THANK YOU